How Should CFD-AI Benchmarks Be Reviewed? Validation Axes, Splits, Failure Modes, and Reproducibility

Skill-guided sample

Abstract

CFD-AI benchmark reviews should not be written as resource catalogues. A benchmark supports a manuscript claim only when its workflow role, validation axis, split design, failure mode, and reproducibility metadata match the claim being made. This mini-review distinguishes datasets, tools, benchmark suites, curated lists, and review claims, then organizes CFD-AI benchmarking around held-out time, Reynolds or parameter regime, geometry, mesh, boundary and initial conditions, cross-solver transfer, coupled-solver deployment, physical diagnostics, and data logistics. The contribution is a reviewer-facing taxonomy and checklist for judging benchmark evidence while blocking unsupported deployment or ranking claims unless direct evidence is supplied.

1 Introduction

CFD-AI manuscripts often use benchmark names as evidence shortcuts. That shortcut is unsafe. A broad PDE dataset can test operator learning under selected PDE families, a PINN library can support a workflow, an aerodynamic dataset can test geometry-conditioned prediction, and a curated list can improve literature coverage. None of these, by itself, proves deployment in a CFD solver or generalization to unseen geometries, meshes, boundary conditions, or regimes.

This review treats benchmark design as a claim-evidence problem. The first question is not which model is best, but which claim the benchmark can support. A random snapshot split supports a different claim from a held-out geometry split; same-grid field error supports a different claim from mesh-transfer conservation or force-coefficient agreement; offline inference timing supports a different claim from coupled-solver wall-clock speedup. Missing source details, license terms, solver settings, dataset sizes, and leaderboard numbers are therefore marked as TODO rather than filled in from memory.

2 Source Boundaries and Benchmark Objects

A review should first separate the object under discussion. Datasets, tools, benchmark suites, curated lists, and review claims have different evidentiary status.

3 Workflow-First Benchmark Taxonomy

Benchmark reviews should classify CFD-AI tasks by workflow role before naming model families. A surrogate used for design screening, a closure term coupled to RANS or LES, a reconstruction model for sparse measurements, and a flow-control policy fail in different ways.

Object	Example role	What it can support	Limitation / TODO
Benchmark suite	PDEBench-like PDE tasks	Split and model-family comparison within supplied PDE cases	TODO: exact task list, versions, scores, licenses
Tool/library	DeepXDE-like PINN tooling	Workflow feasibility for residual/BC/IC experiments	A tool feature is not validation evidence
Engineering dataset	DrivAerNet-like aero data	Geometry-conditioned fields or force prediction under stated splits	TODO: version, license, mesh policy, split leakage
Physics collection	The Well-like datasets	Large spatiotemporal benchmark logistics and rollout tests	Not all entries are CFD; classify governing physics
Curated list	ML-fluid resource list	Coverage discovery and source-scope planning	List inclusion is not benchmark quality

Table 1: Source-boundary table. The table supports the claim that resource type determines what evidence may be inferred, and what must remain TODO.

4 Split Design and Validation Axes

The phrase "generalization" is too coarse for CFD-AI benchmark review. Held-out axes should be separated because each supports a different manuscript claim.

5 Failure Modes and Reviewer Traps

Reviewer-risk language should be rewritten before submission. Unsupported claims are not made safer by smoother prose.

6 Reproducibility, Licensing, and Data Logistics

Benchmark reuse depends on more than model architecture. A review should ask for dataset version, source generation procedure, solver and mesh details, preprocessing, split files, seeds, hardware, runtime, license, storage format, and scripts needed to reproduce metrics. Missing values should be recorded as TODO rather than guessed. For public claims about PDEBench-like, DeepXDE-like, DrivAerNet-like, or The Well-like resources, exact citation metadata, dataset sizes, license terms, and current benchmark numbers must be verified against the current source.

7 Evidence-Linked Figure Placeholders

Figure placeholder 1: Benchmark landscape map. Claim supported: CFD-AI benchmarks should be grouped by workflow role and validation axis, not by resource popularity.

Figure 1: Benchmark landscape map placeholder.

Workflow role	Validation axis	Failure mode	Reproducibility minimum
Data-driven surrogate	Held-out time, regime, geometry, mesh, QoI	Random snapshot split overstates transfer	Solver, grid, split, metrics, runtime
Physics-informed surrogate	Residual, BC/IC transfer, physical diagnostic	Residual loss mistaken for diagnostic agreement	PDE form, nondimensionalization, residual weights
Solver acceleration	Coupled-solver stability and cost	Offline accuracy fails inside solver loop	Coupling interface, CFL/time step, wall-clock accounting
Turbulence closure	A priori and a posteriori evidence	Closure-term fit worsens mean flow after feedback	Target definition, invariance, solver integration, UQ
Reconstruction	Noise, sparsity, spectra, derivative QoIs	Visual agreement hides spectral/gradient error	Measurement model, noise level, train/test regimes
Aerodynamic design	Held-out geometry family/topology	Near-duplicate geometry leakage	Geometry parameterization, force integration, split by design
Flow control	Closed-loop stability, delay, energy budget	Offline reward does not transfer to plant	Observation/action limits, seeds, baseline controller
Multiphase/reacting flow	Interface/regime transfer, balance checks	Smooth field metrics miss topology failure	Phase variables, closure assumptions, balance checks

Table 2: Workflow and validation taxonomy. The table supports the claim that benchmark evidence must be matched to CFD workflow role and expected failure mode.

Figure placeholder 2: Split-design decision tree. Claim supported: random, temporal, parameter, geometry, mesh, BC/IC, cross-solver, and coupled-solver splits validate different statements.

Figure 2: Split-design decision tree placeholder.

8 Research Agenda

A useful CFD-AI benchmark agenda should prioritize: (i) published split files that isolate time, regime, geometry, mesh, BC/IC, and solver-coupling claims; (ii) physical diagnostics beyond field MSE, including spectra, balance checks, forces, stresses, dissipation, uncertainty calibration, and long-rollout drift; (iii) reproducibility metadata for solver, mesh, preprocessing, license, seeds, hardware, and runtime; and (iv) negative results or failure-mode reports that show where models should not be used.

9 Conclusion

Benchmark reviews should make CFD-AI claims harder to overstate. The correct unit of review is not the resource name, but the evidence path from resource type to validation axis, split design, failure mode, and reproducibility. When that path is incomplete, the defensible output is TODO, not a polished claim.

Axis	Claim it may support	Evidence needed	Unsafe shortcut
Held-out time	Bounded rollout behavior	rollout protocol, drift metric, failure horizon	one-step loss implies stability
Regime / parameter	transfer within stated nondimensional range	train/test ranges, regime labels	random mixing implies regime transfer
Geometry/topology	design-space transfer	family-level split, leakage check, force/field diagnostics	near-duplicate shapes imply new geometry
Mesh/resolution	discretization transfer	resolution ladder, projection, balance/QoI drift	same-grid error implies mesh independence
BC/IC	scenario transfer	boundary definitions, sensitivity, residuals	residual term implies boundary transfer
Cross-solver/fidelity	portability	solver/fidelity definitions, uncertainty	one solver dataset implies deployment evidence
Coupled solver	numerical usefulness	interface, CFL/time step, residual behavior, fallback policy	neural inference timing implies CFD acceleration

Table 3: Validation-axis table. The table supports the claim that each split design validates a different kind of CFD-AI statement.

Figure placeholder 3: Failure-mode and reproducibility funnel. Claim supported: benchmark evidence weakens when source scope, split design, failure mode, license, and reproduction metadata are missing.

Figure 3: Failure-mode and reproducibility funnel placeholder.

References

[1] TODO: verified PDEBench citation.

[2] TODO: verified DeepXDE citation.

[3] TODO: verified DrivAerNet or DrivAerNet++ citation and license/version.

[4] TODO: verified The Well citation and dataset-scope note.

[5] TODO: verified curated ML-fluid resource-list citations.

Trap	Safer rewrite	Evidence required
"comprehensive CFD-AI benchmark"	benchmark covers the supplied domains only	source-scope table and TODO gaps
"top-ranked model"	compared under the supplied same-data/same-split setting	verified leaderboard or same-scope baseline
"diagnostically verified PINN"	residual/BC terms were included; diagnostic agreement remains TODO	balance checks, spectra, forces, residuals
"deployment-speed CFD acceleration"	offline inference cost was measured; coupled speedup remains TODO	wall-clock solver accounting
"new-geometry transfer"	evaluated on the specified held-out geometry split	leakage check, force/field diagnostics
"closure ready for solver use"	a posteriori coupled-solver evidence is required	stability, residual, QoI, UQ evidence

Table 4: Reviewer-trap rewrite table. The table supports the claim that benchmark reviews must convert hype terms into evidence-bounded statements.